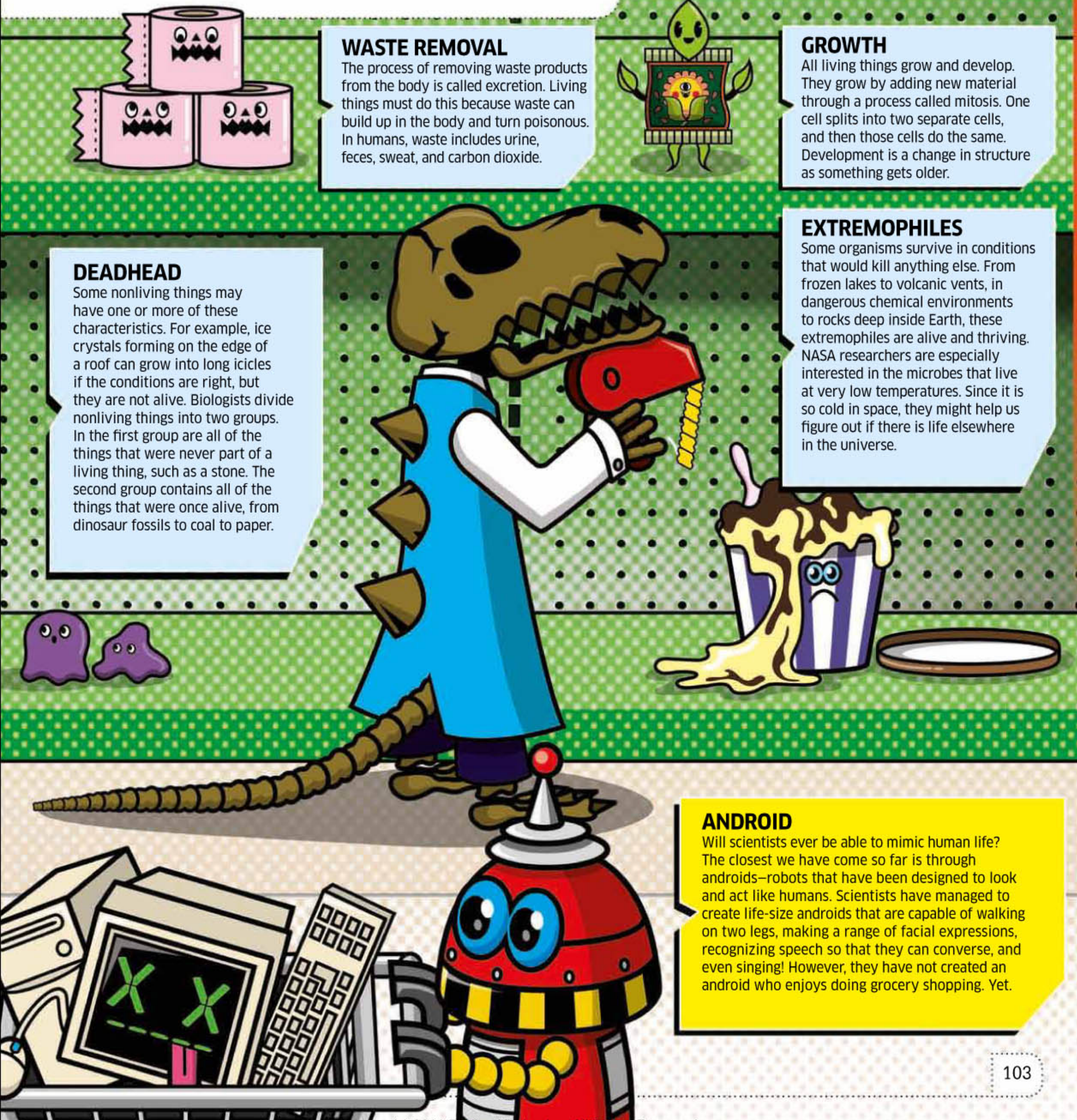
WHAT IS LIFE ANYWAY?

This seems like a silly question. After all, if you are reading this, you must be living, right? We all know what it means to be alive: you move, eat, breathe, go to the supermarket, and so on. Yet it isn't very easy to describe what life itself is, and it is not always easy to tell whether or not something is alive. This question has puzzled philosophers and had scientists scratching their heads for centuries. Let's shop around for an answer. Are you ready to dash to the market?



WASTE REMOVAL

The process of removing waste products from the body is called excretion. Living things must do this because waste can build up in the body and turn poisonous. In humans, waste includes urine, feces, sweat, and carbon dioxide.

GROWTH

All living things grow and develop. They grow by adding new material through a process called mitosis. One cell splits into two separate cells, and then those cells do the same. Development is a change in structure as something gets older.

DEADHEAD

Some nonliving things may have one or more of these characteristics. For example, ice crystals forming on the edge of a roof can grow into long icicles if the conditions are right, but they are not alive. Biologists divide nonliving things into two groups. In the first group are all of the things that were never part of a living thing, such as a stone. The second group contains all of the things that were once alive, from dinosaur fossils to coal to paper.

EXTREMOPHILES

Some organisms survive in conditions that would kill anything else. From frozen lakes to volcanic vents, in dangerous chemical environments to rocks deep inside Earth, these extremophiles are alive and thriving. NASA researchers are especially interested in the microbes that live at very low temperatures. Since it is so cold in space, they might help us figure out if there is life elsewhere in the universe.

ANDROID

Will scientists ever be able to mimic human life? The closest we have come so far is through androids—robots that have been designed to look and act like humans. Scientists have managed to create life-size androids that are capable of walking on two legs, making a range of facial expressions, recognizing speech so that they can converse, and even singing! However, they have not created an android who enjoys doing grocery shopping. Yet.